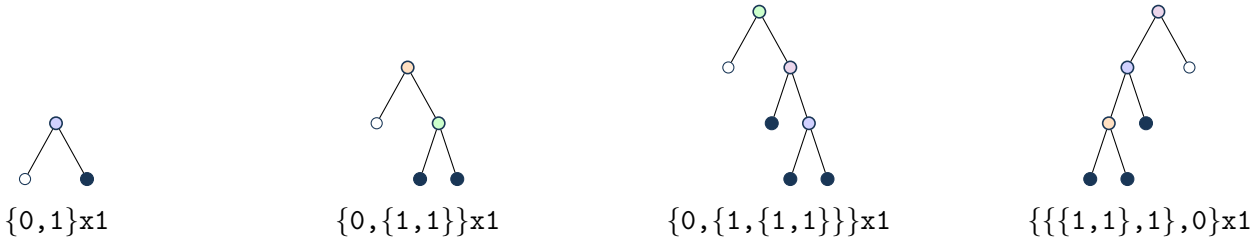


Definition and first terms

$A(x) = A_{\text{parent}}(x)^2$, $A_{\text{parent}}(x) = A_{120588}(x)$, with the origin-normalized solution.
Normalization/grammar: ordered forest of 2 parent A120588 typogeometries.
 $a(0), \dots, a(7) = 1, 2, 3, 6, 15, 42, 126, 396$.
Kernel used throughout: $\rho(u) = u(1 - 1u^1)$, so $x = \rho(T(x))$.

Typogeometry



forest slots are explicit; 0 denotes an empty slot. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 1.

Coefficient and contour extraction

$$p(n) = [x^n]A_{\text{parent}}(x),$$
$$a(n) = \sum_{\substack{n_1, \dots, n_2 \geq 0 \\ n_1 + \dots + n_2 = n}} \prod_{j=1}^2 p(n_j).$$

$$[x^n]A(x)^2 = \frac{2}{2\pi i n} \oint_{\gamma} \frac{(1+u)^1 du}{u^n D(u)^n}.$$

Recurrence

$$\sum_{r=0}^2 P_r(n)a(n+r) = 0, \quad n \geq 1, \text{ with}$$
$$P_0(n) = 0$$
$$P_1(n) = -4n - 2$$
$$P_2(n) = n + 2$$

Differential equation

$$(2x)A(x) + (-4x^2 + x)A'(x) = 2x^2 + 4x.$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n, u) = N(n, u)/(-u^2 + u)^1.$$

$$\text{Telescoping identity: } \sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n, u)H_n(u)).$$

Matrix dimensions: 4×4 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.